

What is Programming? * Computer ko instructions (hidayat) dena takay wo koi specific kaam kar sakay.

- Programming languages (jaise JavaScript aur Python) computer se baat karne ka zariya hain.
- Data types:
 - a. String: ""
 - b. Number: 672389
 - c. Boolean: True or False

JAVASCRIPT OUTPUT:

- Hum browser ke developer console mein result dekhne ke liye console.log () ka istemal karte hain.
- Console is the building function of javascript.

JAVASCRIPT BASIC:

1. Javascript is backent and fronkent and all rounder.
2. javascript is language used to make web pages interactive.
3. it runs in all modern web browsers and can also run on servers using node.js.
4. it works together with HTML and CSS.
5. javascript is case-sensitive.
6. if and else are used to make decisions.
7. Javascript Variables:
 - a. Const
 - b. Let
 - c. Var

PYTHON (OUTPUT):

- Python mein screen par kuch bhi dikhane ke liye print () function use hota hai. Isme koi semicolon (;) nahi lagta.
- Python mein hum jo likhtein hein use print statement bhi kehtein hain.

PYTHON BASIC:

1. Python mein variable banane ke liye let ya var ki zaroorat nahi hoti. Aap direct naam likh kar value de dete hain.
2. String (Text): Quotes ke andar likha jata hai.
3. Number (Integer/Float): Bina quotes ke simple digits.

Feature	JavaScript 	Python 
Output	<code>console.log("Hello");</code>	<code>print("Hello")</code>
Variable	<code>let x = 10;</code>	<code>x = 10</code>
Code Block	Curly Brackets { } use hote hain	Spaces (Indentation) use hoti hai
Line End	Semicolon ; lagaya jata hai	Koi semicolon nahi lagta

JavaScript Syntax	Python Syntax
<code>javascript let age = 16; let name = "Maheen";</code>	<code>python age = 16 name = "Maheen"</code>

ALERT

- `alert ()` JavaScript ka ek function hai jo browser par ek pop-up message box dikhata hai.
- Kaam: User ko warning ya notification dena.

- Rule: Jab tak user "OK" par click nahi karta, wo page par kuch aur nahi kar sakta.

VARIBLES FOR STRINGS

- **Variables for Strings** ka matlab hai ek aisa container ya box banana jo **text (lafz ya jumle)** ko store kare.
- String ko hamesha **Quotation Marks** ("" ya ") ke andar likha jata hai.

VARIBLES FOR NUMBERS

- **Variables for Numbers** ka matlab hai ek aisa container banana jo **mathematical values (ginti ya digits)** ko store kare.
- Numbers ke sath **quotation marks ("") nahi lagaye jaate**, warna computer use text samajh lega.

VARIABLE NAMES LEGAL AND ILLEGAL

Legal Variable Names (Allowed)	Illegal Variable Names (Not Allowed)
let my Name = "Maheen"; (CamelCase is good)	let 1name = "Maheen"; (Number se shuru nahi ho sakta)
let user age = 16; (Underscore allowed hai)	let user age = 16; (Space allowed nahi hai)
let \$price = 100; (Dollar sign allowed hai)	let user@id = 123; (Special character allowed nahi hai)

MATH EXPRESSIONS FAMILIAR OPERATORS

- Familiar **Operators**: Yeh wo Aam symbols hain jo computer ko +, -, *, / karne ke liye bataati hain.
- Dono JavaScript aur Python mein yeh operators bilkul **same** tareeqay se kaam karte hain.

MATH EXPRESSIONS UNFAMILIAR OPERATORS

- Unfamiliar **Operators**: Yeh wo operators hain jo programming mein khas kaamon ke liye use hote hain (jaise remainder nikalna ya value ko 1 number barhana).
- Modulus (%): Yeh divide karne ke baad bacha hua hissa (Remainder) batata hai.
- Increment (++): Kisi bhi number mein 1 jama (add) kar deta hai (*Sirf JavaScript mein*).

Math expression

Eliminating ambiguity

- **Decrement (--)**: Kisi bhi number mein se 1 nafi (subtract) kar deta hai (*Sirf JavaScript mein*).
- **BODMAS / PEMDAS Rule**: Computer math expressions ko ek khaas order (tarteeb) mein solve karta hai.
- **Order of Preference**: 1. Brackets () (Pehle brackets ke andar ka hissa solve hoga) 2. Multiply * aur Divide / 3. Plus + aur Minus -
- **Rule**: Agar aap chahte hain ke koi cheez pehle solve ho, toh uske aas-paas **Brackets ()** laga dein.

CONCATENATING TEXT STRINGS

- **Concatenation**: Do ya do se zyada strings (text) ko aapas mein jor kar ek bara sentence banana.
- The **Operator**: Is kaam ke liye hum **Plus (+)** sign ka istemal karte hain.

- Note: Jab do numbers ke darmiyan + lagayein toh wo maths karta hai, lekin do strings ke darmiyan + lagayein toh wo unhein jor deta hai.

PROMPT

- **prompt () Function:** Yeh browser mein ek pop-up box kholta hai jahan user kuch type kar sakta hai.
- **Purpose:** Iska maqsad user se input (jaise unka naam, age, ya koi number) lena hota hai.
- **Note:** Prompt user se jo bhi data leta hai, computer use hamesha ek **String (text)** samajhta hai.

IF STATEMENT

- **if Statement:** Yeh tab use hota hai jab hum chahte hain ke koi code *sirf tabhi* chale jab koi condition **True (sahi)** ho.
- **How it works:** Computer condition check karta hai; agar condition sahi hai toh code chalta hai, warna computer us code ko skip (Chor) deta hai.

COMPARISON OPERATORS

Operator	Meaning	Example Code	Result
>	Greater than (Bara hai)	5 > 3	True
<	Less than (Chota hai)	10 < 2	False
>=	Greater than or equal to	16 >= 16	True
==	Equal to (Barabar hai)	"abc" == "abc"	True
!=	Not equal to (Barabar nahi hai)	5 != 5	False

IF...ELSE AND ELSE IF STATEMENTS

- **if...else Statement:** Agar if wali condition galat (False) ho jaye, toh computer automatically else wala code chala deta hai (Yeh ek backup plan ki tarah hota hai).
- **else if Statement:** Agar आपको pehli condition ke baad **ek aur nayi condition** check karni ho, toh beech mein else if lagaya jata hai.
- **Rule:** Computer upar se shuru karta hai; jo bhi condition pehle sahi (True) milti hai, sirf uska code chalta hai aur baqi sab skip ho jata hai.